

LAB OBSERVATION

EXPERIMENT-11

Implement a simple top-down parser for context-free grammars using python.

CODE:

```
[1] ✓ 22s
1 import nltk
2 from nltk import CFG
3 from nltk.parse import RecursiveDescentParser
4
5 grammar = CFG.fromstring("""
6 S -> NP VP
7 NP -> 'Ram'
8 VP -> V NP
9 V -> 'likes'
10 """)
11
12 parser = RecursiveDescentParser(grammar)
13
14 sentence = input("Enter sentence: ").split()
15
16 for tree in parser.parse(sentence):
17     print(tree)
```

Variables Terminal 1:52 PM Python 3

C 36°C Mostly sunny

OUTPUT:

```
... Enter sentence: Ram likes Ram
(S (NP Ram) (VP (V likes) (NP Ram)))
```

Variables Terminal 1:52 PM Python 3

C 36°C Mostly sunny